P510/3 PHYSICS PRACTICAL

CODE	MARKING GUIDE DESCRIPTOR	MARK	COMMENT
	QUESTION 1 (40 marks)		
A_1	Letting t or using 20 T……	½	
A_2	Value of 2 L_2 (with unit) ……	1	
		1½	
B_0	All values of d in first column, order of table labels and closed table. {d, L_1, 2Θ, t or 20T, T, T^2, Θ, cos Θ, $L_1 cos\ \Theta$, ($L_1 cos\ \Theta$ + 2 L_2)} ……	1½	
B_1	Columnar table labelled: (d, L_1, 2Θ, t or 20T, T, T^2, Θ, cos Θ, $L_1 cos\ \Theta$,($L_1 cos\ \Theta$ + 2 L_2) ;(**correct symbol @1/2**) .	5	
B_2	Units in brackets: **m, m, ^{0}, s, s, s^2, ^{0}, -, m, m;** (**correct symbols @ ½**)……	5	
B_3	Values of L_1, **2dp decreasing** (1.10, 0,90, 0.80, 0.70, 0.60, 0.50) ……	$\frac{1}{2}$	
B_4	Values of 2Θ, **0dp decreasing** (105^0-27^0) (**@1/2**) ……	3	

Let **t** be time for 20 oscillations.

d(m)	L_1(m)	2Θ(^{0})	**t**(s)
0.90	1.10	$105-115$	$29.5-31.5$
0.70	0.90	$75-85$	$32.5-34.5$
0.60	0.80	$64-74$	$33.0-35.0$
0.50	0.70	$53-63$	$34.5-36.5$
0.40	0.60	$38-48$	$34.5-36.5$
0.30	0.50	$27-37$	$35.0-37.0$
			-Values increasing by 0.5-1.0 s **-Allow repetition once**

B_5	Values of **t, 1dp or 2dp increasing** (29.5-37.0) (**@1/2**) ……	3	
B_6	Correctly calculated values of T **2dp or 3dp** (**@ pair, ½)** ……	1½	
B_7	Correctly calculated values of T^2 **2dp or 3dp** (**@ pair, ½)** ……	1½	
B_8	Correctly calculated values of Θ, **1dp** (**@ pair, ½)** ……	1½	
B_9	Correctly calculated values of cos Θ, **3dp** (**@ pair, ½)** ……	1½	
B_{10}	Correctly calculated values of $L_1 cos\ \Theta$ **2dp** (**@ pair, ½)** ……	1½	
B_{11}	Correctly calculated values of ($L_1 cos\ \Theta$ + 2L_2) **2dp** (**@ pair, ½)** ……	1½	
		27	

C_1	Title: ***A graph of*** T^2 ***against*** ($L_1 cos\ \theta + 2L_2$) …….	**½**	
C_2	Axes perpendicular with arrows at end, correctly labelled; **VA:** T^2 **HA:** $L_1 cos\ \theta + 2L_2$) [correct symbols] @ ½ …….	**½ + ½**	
C_3	Uniform scales, with starting values indicated, marked at least three times @ ½ ……	**½ + ½**	
C_4	Correctly plotted points [no shading] @ ½ …… [**If axes not labelled or reversed, no mark**]	**3**	
C_5	Line of best fit provided 4 points correctly plotted …….	**½**	
C_6	Indication of slope covering all the plotted points ……. [*If right-angled triangle, must touch line of best fit*]	**½**	
C_7	Coordinates for slope correctly read @ axis ½ …….	**½ + ½**	
C_8	Correctly calculated $\boldsymbol{S} = (1.04 - 2.20)$, **_1dp or 2dp;** unit: $s^2\ m^{-1}$…….	**1 + ½**	
C_9	Correctly calculating the value of $\mathbf{g} = (\ 9.0 - 11.0\ )$ **1dp or 2dp**; unit: $m\ s^{-2}$ ……	**1 + ½**	
C_{10}	Two correct sources of error and their correct precautions **@ pair ½** ……	**1**	
		11½	

TOTAL MARKS = 40

CODE	MARKING GUIDE DESCRIPTOR	MARK	COMMENT
	QUESTION 2 (40 marks)		
A_1	Value of $d = (3.90 - 4.40)$, **2dp;** unit: **cm**	**1 + ½**	
A_2	Correctly calculated value of $t = (d - \sqrt{2})$, **1dp**	**½**	
A_3	Correctly calculated value of $m = (d^2 - 1)$, **odp**	**½**	
A_4	Correctly calculated value of m^2, **odp**	**½**	
		3	
B_0	All values of i in first column, order of table labels and closed table. $\{i, a, (a-t), (a-t)^2, a^2, (a^2+\sqrt{2}), \left(m^2+(a^2+\sqrt{2})\right), y, \sin i,\ sin^2 i\}$	**1½**	
B_1	Columnar table labelled: $\{i, a, (a-t), (a-t)^2, a^2, (a^2+\sqrt{2}), \left(m^2+(a^2+\sqrt{2})\right), y, \sin i,\ sin^2 i\}$; (**correct symbol @1/2**)	**5**	
B_2	Units in brackets: **⁰, cm, cm, cm², cm², cm², cm², −or cm^{-2}, −, −;** (**correct symbols @ ½**).....	**5**	
B_3	Values of a, **1dp decreasing** $(2.2 - 0.7)$ (**@ 1**).....	**6**	

$i(^0)$	a(cm)							$sin\ i$	$sin^2 i$
15	$1.8 - 2.2$							0.259	0.067
20	$1.6 - 2.0$							0.342	0.117
25	$1.3 - 1.7$							0.423	0.179
30	$1.2 - 1.6$							0.500	0.250
35	$0.9 - 1.1$							0.574	0.329
40	$0.7 - 1.1$							0.643	0.413

B_4	Correctly calculated values of $(a-t)$, **1dp** (**any three, ½)**	**1**	
B_5	Correctly calculated values of $(a-t)^2$ **1dp** (**any three, ½)**	**1**	
B_6	Correctly calculated values of a^2 **1dp** (**any three, ½)**	**1**	
B_7	Correctly calculated values of $(a^2+\sqrt{2})$ **1dp** (**any three, ½)**	**1**	
B_8	Correctly calculated values of $\left(m^2+(a^2+\sqrt{2})\right)$ **1dp** (**any three, ½)**	**1**	

B_9	Correctly calculated values of y **3dp** (**any three, ½)**	**1**
B_{10}	Correctly calculated values of $\sin i$ **3dp** (**any three, ½)**	**1**
B_{11}	Correctly calculated values of $sin^2 i$ **3dp** (**any three, ½)**	**1**
		25½
C_1	Title: ***A graph of*** $sin^2 i$ ***against*** y.....	**½**
C_2	Axes perpendicular with arrows at end, correctly labelled; **VA:** $sin^2 i$ **HA:** y [**correct symbols**] **@ ½**	**½ + ½**
C_3	Uniform scales, with starting values indicated, marked at least three times @ ½	**½ + ½**
C_4	Correctly plotted points [no shading] @ ½ ..... [**If axes not labelled or reversed, no mark**]	**3**
C_5	Line of best fit provided 4 points correctly plotted	**½**
C_6	Indication of slope covering all the plotted points ..... [*If right-angled triangle, must touch line of best fit*]	**½**
C_7	Coordinates for slope correctly read @ axis ½	**½ + ½**
C_8	Correctly calculated $\boldsymbol{s = (32 - 45)}$, **0dp;** unit: no unit or cm^2.....	**1 + ½**
C_9	Correctly calculating the value of $\mathbf{n = (5.7 - 6.7)}$; unit: $\boldsymbol{no\ unit}$ ..... $\boldsymbol{Note}$: $\boldsymbol{The\ expression\ would\ have\ been\ s = (dn)^2}$ **not** $\boldsymbol{s = n^2}$	**1 ½**
C_{10}	Two correct sources of error and their correct precautions **@ pair ½**	**½ + ½**
		11½

TOTAL MARKS = 40

CODE	MARKING GUIDE DESCRIPTOR	MARK	COMMENT
	QUESTION 3 (40 marks)		
A_1	Value of $D = (0.28 - 0.34)$, **2dp;** unit: **mm**	**1 + ½**	
A_2	Value of $I = (0.68 - 0.88)$, **2dp;** unit: **A**	**½ + ½**	
A_3	Value of $V = (2.00 - 2.40)$, **2dp or 1dp;** unit: **V**	**½ + ½**	
A_4	Correctly calculated value of $\boldsymbol{\varphi} = (6.1 - 9.1)\ \text{x}\ 10^{-8}$, unit: m^2	**1 + ½**	
A_5	Correctly calculated value of $\boldsymbol{f_1} = (4.7 - 5.4)\ \text{x}\ 10^{-7}$, unit: $\Omega\, m$	**2½ + ½**	
		8	
B_0	All values of $\boldsymbol{y}$ in first column, order of table labels and closed table. $\{\boldsymbol{y},\ \boldsymbol{l_1}, \boldsymbol{l_1},\ \boldsymbol{l_2},\ \boldsymbol{l_2}, \frac{l_1}{l_2},\ \frac{\boldsymbol{y}}{\boldsymbol{R}}\}$	**1½**	
B_1	Columnar table labelled: $\{\boldsymbol{y},\ \boldsymbol{l_1}, \boldsymbol{l_1},\ \boldsymbol{l_2},\ \boldsymbol{l_2}, \frac{l_1}{l_2},\ \frac{\boldsymbol{y}}{\boldsymbol{R}}\}$ (**correct symbol @ ½**)	**3½**	
B_2	Units in brackets: **m, cm, m, cm, m,** $-$**,** $\boldsymbol{\Omega^{-1}\, m}$**;** (**correct symbols @ ½**).....	**3½**	
B_3	values of $\boldsymbol{l_1}$, **1dp increasing in cm** (any three **½**)..... (Difference between **first and last values** be: $\boldsymbol{31.5 - 36.5}$) (Difference between **successive values** be $\boldsymbol{3.0 - 9.5}$**)**	**1**	
B_4	Values of $\boldsymbol{l_1}$, **3dp increasing in m** (**@ ½**) ..... (Difference between **first and last values** be: $\boldsymbol{0.315 - 0.365}$) (Difference between **successive values** be $\boldsymbol{0.030 - 0.095}$)	**3**	
B_5	Values of $\boldsymbol{l_2}$, **1dp decreasing in cm** (any three **½**)..... (Difference between **first and last values** be: $\boldsymbol{31.5 - 36.5}$) (Difference between **successive values** be $\boldsymbol{3.0 - 9.5}$**)**	**1**	
B_6	Values of $\boldsymbol{l_2}$, **3dp decreasing in m** (**@ ½**) ..... (Difference between **first and last values** be: $\boldsymbol{0.315 - 0.365}$) (Difference between **successive values** be $\boldsymbol{0.030 - 0.095}$)	**3**	
B_7	Correctly calculated values of $\frac{l_1}{l_2}$; **3dp** (any three **½**).....	**1**	
B_8	Correctly calculated values of $\frac{\boldsymbol{y}}{\boldsymbol{R}}$; **3dp** (any three **½**).....	**1**	
		18½	

C_1	Title: ***A graph of*** $\frac{y}{R}$ ***against*** $\frac{l_1}{l_2}$ ……	½	
C_2	Axes perpendicular with arrows at end, correctly labelled; **VA:** $\frac{y}{R}$ **HA:** $\frac{l_1}{l_2}$ [**correct symbols**] **@ ½** ……	½ + ½	
C_3	Uniform scales, with starting values indicated, marked at least three times @ ½ ……	½ + ½	
C_4	Correctly plotted points [no shading] @ ½ …… [**If axes not labelled or reversed, no mark**]	3	
C_5	Line of best fit provided 4 points correctly plotted ……	½	
C_6	Indication of slope covering all the plotted points …… [*If right-angled triangle, must touch line of best fit*]	½	
C_7	Coordinates for slope correctly read @ axis ½ ……	½ + ½	
C_8	Correctly calculated $\boldsymbol{S} = (0.10 - 0.20)$, **2dp;** unit: $\Omega^{-1}\, m$ ……	1 + ½	
C_9	Correctly calculating the value of $\boldsymbol{f_2} = (4.7 - 5.4)\ \text{x}\ 10^{-7}$; unit: $\Omega\, m$ ……	1 + ½	
C_{10}	Correctly calculating the value of $\boldsymbol{f} = (4.7 - 5.4)\ \text{x}\ 10^{-7}$; unit: $\Omega\, m$ ……	1 ½ +½	
C_{11}	Two correct sources of error and their correct precautions **@ pair ½** ……	½ + ½	
		13½	

TOTAL MARKS = 40

E N D